

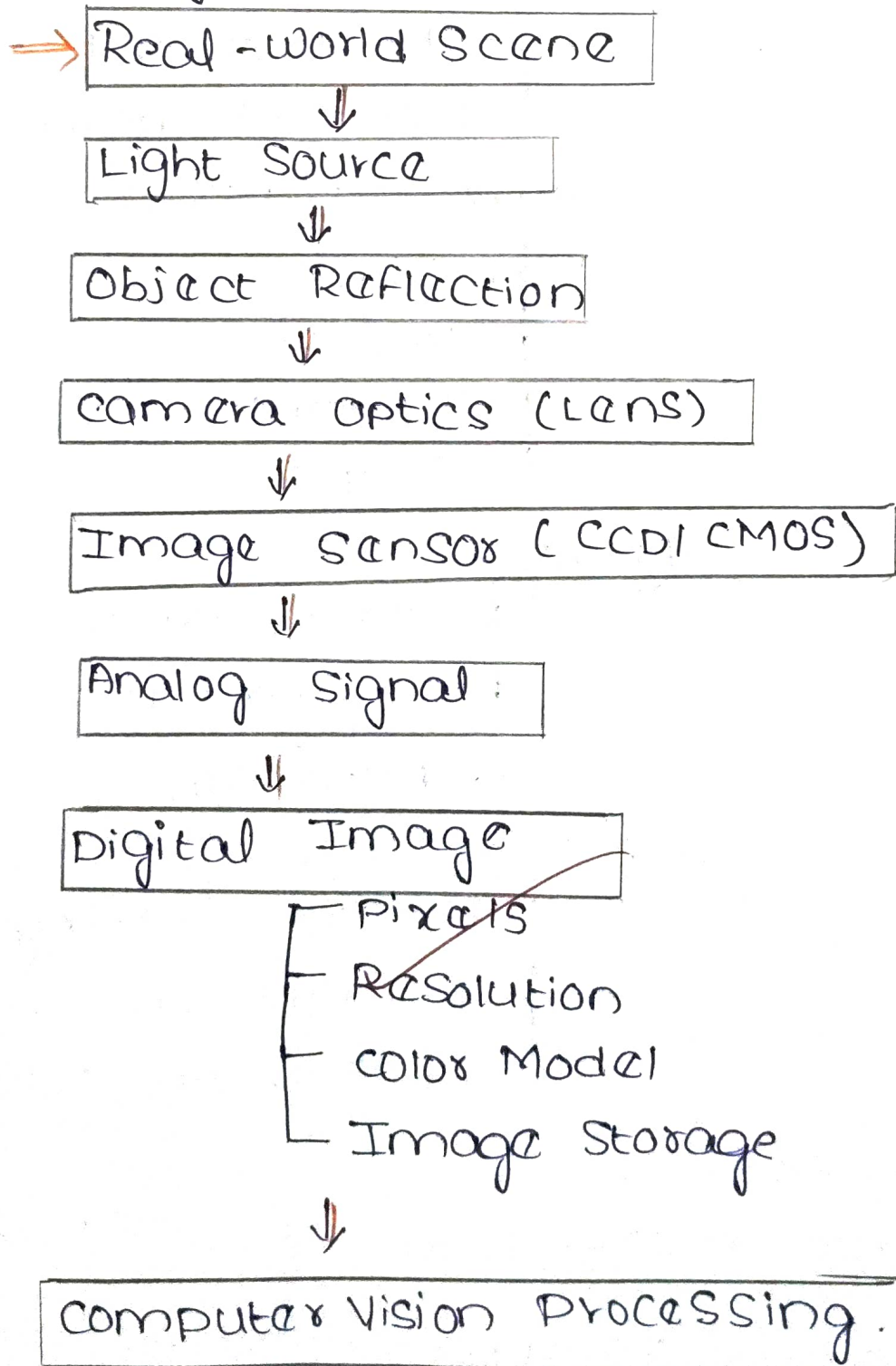
CO1 - AT2 - CONCEPT MAPPING

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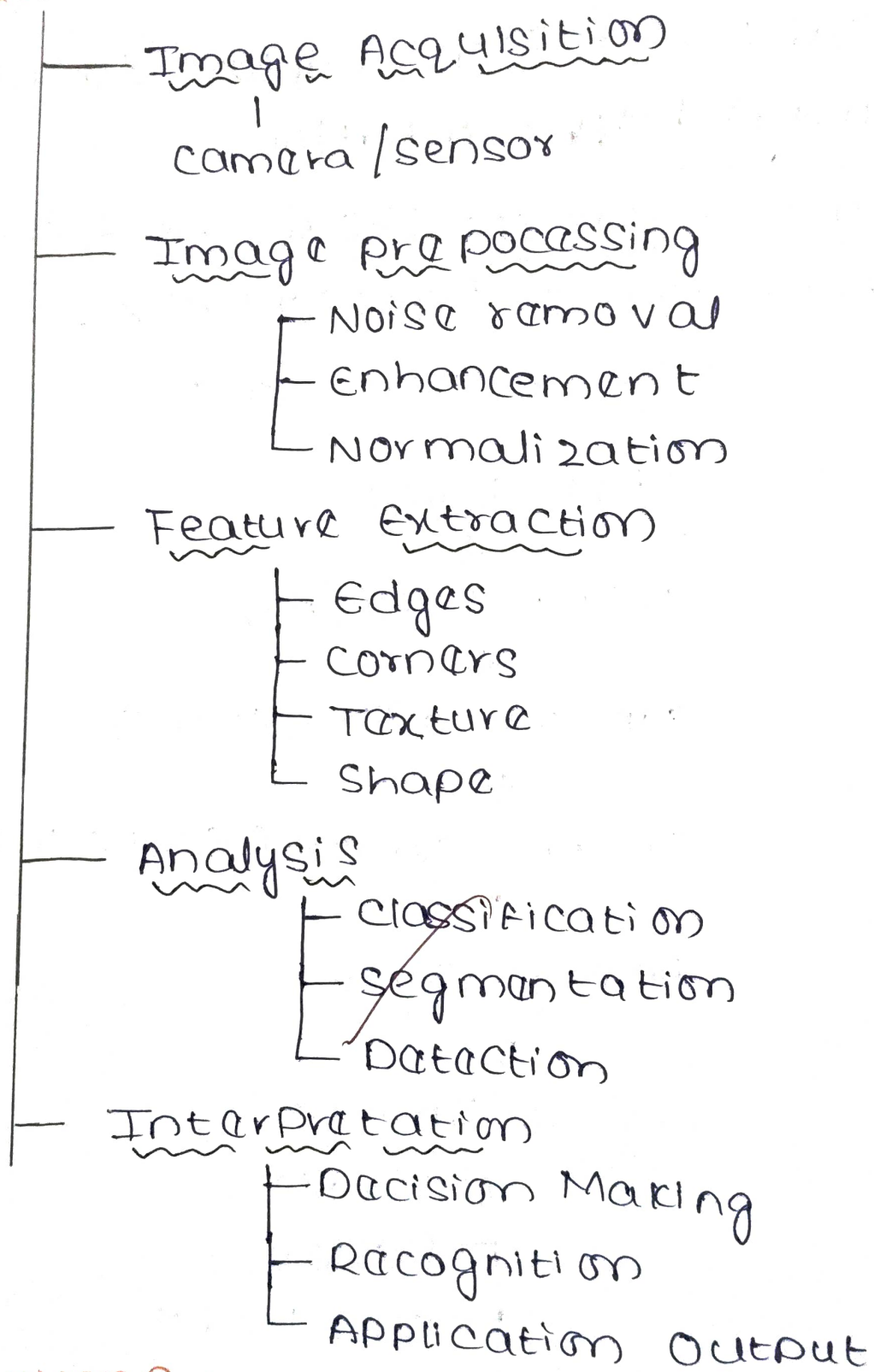
1. Real world Scene capture to Digital Image Representation

A) Central concept : Real-world Scene → Digital image Representation.



2. Major Stages of visual data Handling
central concept : computer vision system.

A. Computer vision system



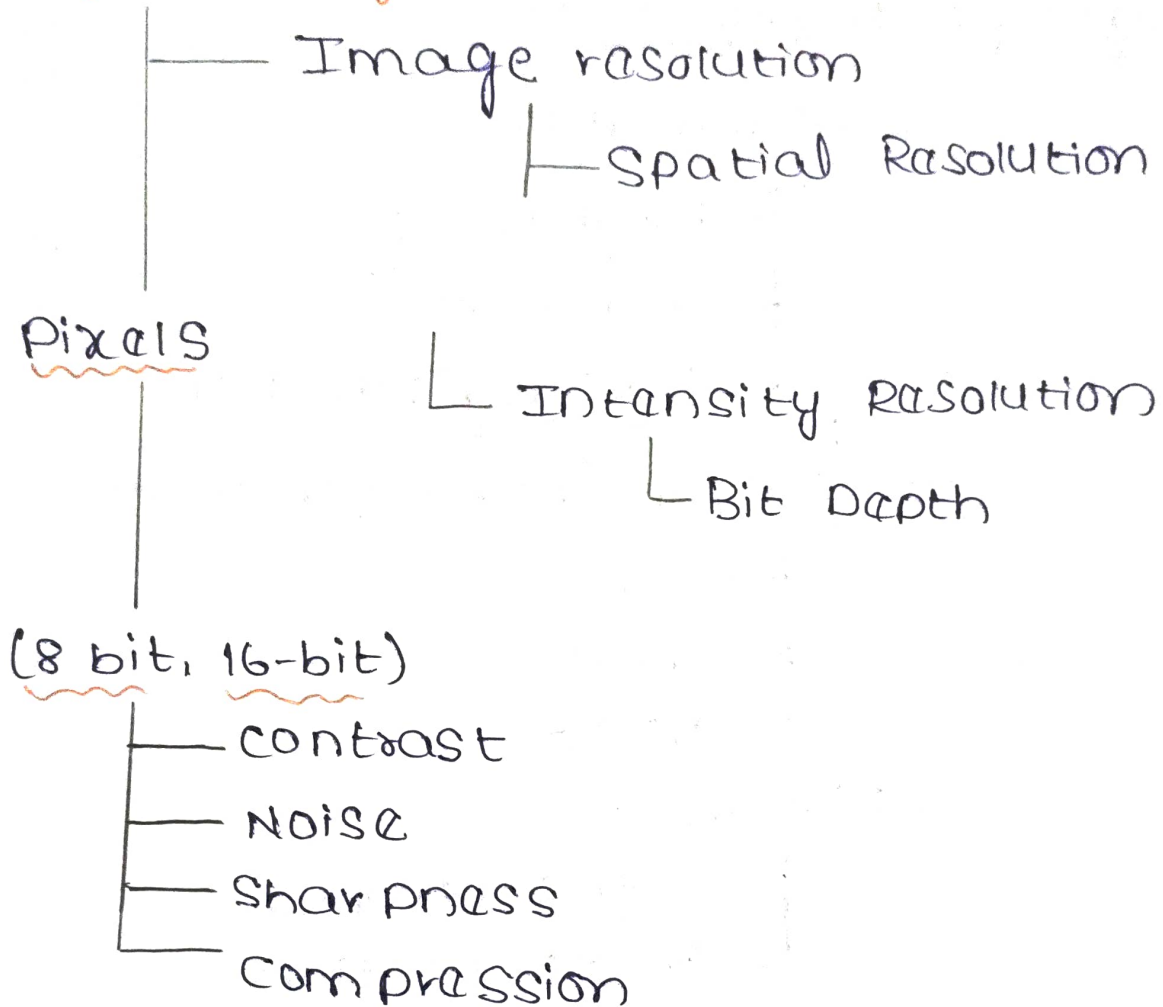
Applications :

- Face recognition
- surveillance
- medical imaging
- Robotics.

Resolution and image quality

Central concept : Image quality

Image quality



Application performance

- Medical imaging
- Satellite Imaging
- Security
- Object Detection

Relationship

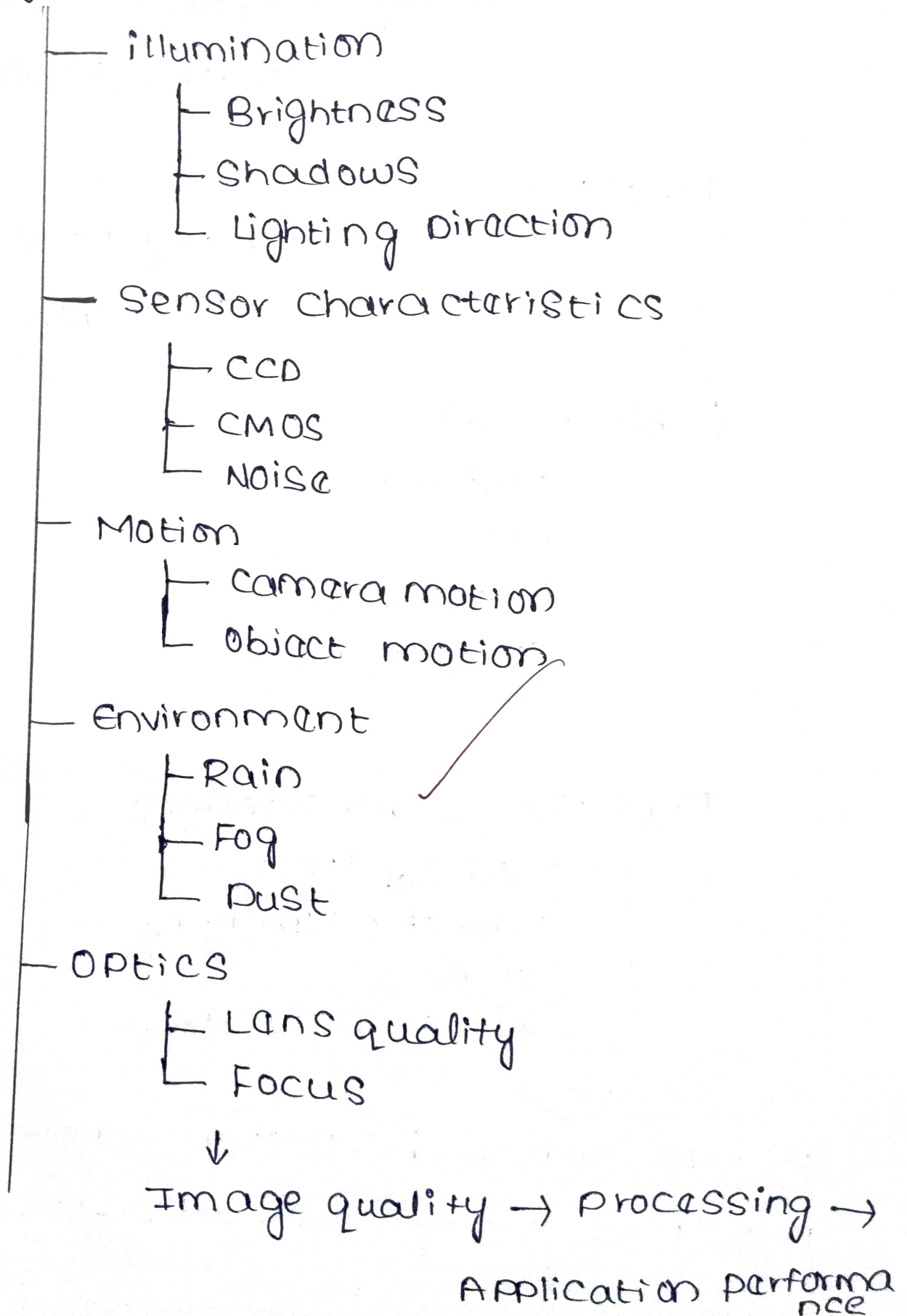
Higher spatial Resolution → More details

Higher intensity → Better Brightness

Factors Affecting Image Acquisition

central concept: Image Acquisition
conditions

Image Acquisition



Computer Vision Capabilities and Applications.

Central concept : Computer vision System

Computer vision :

- Image classification
- Object Detection
- Image Segmentation
- Face Recognition
- Motion Tracking
- OCR
- Pose Estimation



Applications :

- Health care
- Autonomous vehicles
- smart Agriculture
- Manufacturing
- Security
- Retail
- Robotics.

Example Links :

Object Detection → enables → Autonomous

driving

- ⇒ Segmentation → improves → Medical diagnosis
- ⇒ Face Recognition → Supports → Security Systems
- ⇒ OCR → extracts → Text from images.
- ⇒ Motion Tracking → enables → Surveillance.

[Signature]